



1



Mark for Review



The function g is defined by $g(x) = -8x + 60$. What is the value of $g(x)$ when $x = -3$?

- A. 84
- B. 49
- C. 36
- D. $\frac{57}{8}$

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction ($7/2$) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	3 1/2 3 1/2
2/3	2/3 .6666 .6667 0.666	0.66 .66 0.67

2

Mark for Review

A certain blue whale calf weighed 5,600 pounds when it was born, and its weight increased at an average rate of 250 pounds per day for the first n days after birth, where n is an integer. If the calf's weight was greater than 8,850 pounds but less than 9,850 pounds n days after birth, what is one possible value of n ?

3

Mark for Review



$$4x(x - 3)(x + 9) = 0$$

What is one of the solutions to the given equation?

- A. -3
- B. 0
- C. 4
- D. 9

4



Mark for Review



$$y = \frac{x}{3} + 4$$

$$y = -\frac{x}{3} + 40$$

The solution to the given system of equations is (x, y) . What is the value of $2y$?

- A. 4
- B. 36
- C. 40
- D. 44

5



Mark for Review



A hemisphere is half of a sphere. If a hemisphere has a radius of 69 inches, which of the following is closest to the volume, in cubic inches, of this hemisphere?

- A. 10,000
- B. 39,900
- C. 516,000
- D. 688,000



6



Mark for Review



$$f(x) = 18x^2 - 45x + 13$$

The given equation defines the function f . For what value of x does $f(x)$ reach its minimum?

- A. 1.25
- B. 2.25
- C. 13
- D. 45

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction ($7/2$) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	$3\frac{1}{2}$ 3 1/2
$\frac{2}{3}$	2/3 .6666 .6667 0.666	0.66 .66 0.67

7

Mark for Review

An exponential function f is defined by $f(x) = c^x$, where c is a constant greater than 1. If $f(7) = 9 \cdot f(5)$, what is the value of c ?

8



Mark for Review



How many solutions does the equation $10(x + 15) + 5x = 15(x + 10)$ have?

- A. Exactly one
- B. Exactly two
- C. Infinitely many
- D. Zero

Student-produced response directions

- If you find **more than one correct answer**, enter only one answer.
- You can enter up to 5 characters for a **positive** answer and up to 6 characters (including the negative sign) for a **negative** answer.
- If your answer is a **fraction** that doesn't fit in the provided space, enter the decimal equivalent.
- If your answer is a **decimal** that doesn't fit in the provided space, enter it by truncating or rounding at the fourth digit.
- If your answer is a **mixed number** (such as $3\frac{1}{2}$), enter it as an improper fraction ($7/2$) or its decimal equivalent (3.5).
- Don't enter **symbols** such as a percent sign, comma, or dollar sign.

Examples

Answer	Acceptable ways to enter answer	Unacceptable: will NOT receive credit
3.5	3.5 3.50 7/2	$31/2$ $3\ 1/2$
$\frac{2}{3}$	$2/3$.6666 .6667 0.666	0.66 .66 0.67

9

Mark for Review

A line in the xy -plane passes through the points $(7, 1)$, $(0, 4)$, and $(c, 0)$. What is the value of c ?

10



Mark for Review



$$H = 2.403L + 23.31$$

A group of biology students conducted an experiment to study the relationship between a person's femur length and the person's height. The given equation describes the relationship between the length L , in inches, of a student's femur and the student's estimated height H , in inches, for the students in the group. Which of the following is the best interpretation of 2.403 in this context?

- A. The increase in a student's estimated height, in inches, for each increase of 1 inch in the student's femur length
- B. The increase in a student's femur length, in inches, for each increase of 1 inch in the student's estimated height
- C. The increase in a student's femur length, in inches, for each increase of 23.31 inches in the student's estimated height
- D. The estimated height, in inches, of a student whose femur has a length of L inches

11



Mark for Review



$$x(x + 9) - 15 = 0$$

What are the solutions to the given equation?

A. $x = \frac{9 \pm \sqrt{21}}{2}$

B. $x = \frac{9 \pm \sqrt{141}}{2}$

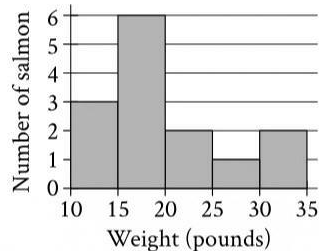
C. $x = \frac{-9 \pm \sqrt{21}}{2}$

D. $x = \frac{-9 \pm \sqrt{141}}{2}$

12



Mark for Review



The histogram summarizes a data set of the weights, in pounds, of 14 salmon. If an additional weight of 85 pounds is added to the original data set to create a new data set of 15 weights of salmon, which of the following measures must be greater for the new data set than for the original data set?

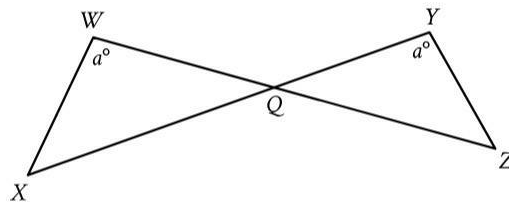
- I. The median
- II. The mean

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II

13

Mark for Review

ABC



Note: Figure not drawn to scale.

In the figure shown, $\overline{WZ}$ and $\overline{XY}$ intersect at point Q , $YQ = 36$, $WQ = 90$, $WX = 70$, and $XQ = 150$. What is the length of $\overline{YZ}$?

- A. 28
- B. 60
- C. 150
- D. 175

Calculator

5

-10 -5 0 5 10

-5

+ ↶ ↷ ⚙ ⌵

powered by

desmos

only one answer.
lower and up to 6 characters
r.
provided space, enter the decimal
provided space, enter it by
enter it as an improper fraction
a, or dollar sign.

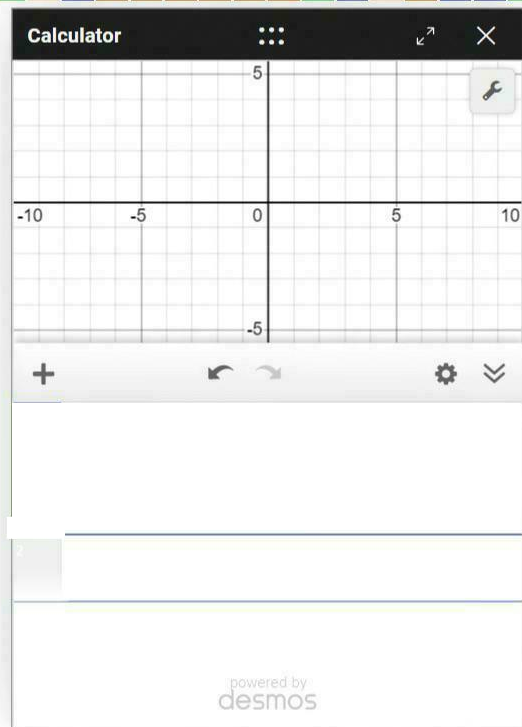
unacceptable: will
NOT receive credit

31/2
3 1/2

2/3	0.66
.6666	.66
.6667	0.67
0.666	

14 Mark for Review

A researcher surveyed undergraduate students, graduate students, and postdoctoral students. The number of undergraduate students surveyed was 6,950% of the number of postdoctoral students surveyed and the number of graduate students surveyed was 35% of the number of undergraduate students surveyed. If there were 4,865 graduate students surveyed, what was the sum of the number of undergraduate students and postdoctoral students surveyed?



15

Mark for Review



The function $T(x) = \frac{22,000 - 6.5x}{1,000}$ gives the estimated air temperature $T(x)$, in degrees Celsius ($^{\circ}\text{C}$), surrounding a hot air balloon at an altitude of x meters. If the estimated air temperature surrounding the hot air balloon is 16°C , which of the following is closest to the altitude, in meters, of the hot air balloon?

- A. 22
- B. 923
- C. 1,375
- D. 5,846

Calculator

only one answer.

wer and up to 6 characters

r.

provided space, enter the decimal

provided space, enter it by

enter it as an improper fraction

a, or dollar sign.

acceptable: will
OT receive credit

31/2
3 1/2

2/3
.6666
.6667
0.666

0.66
.66
0.67

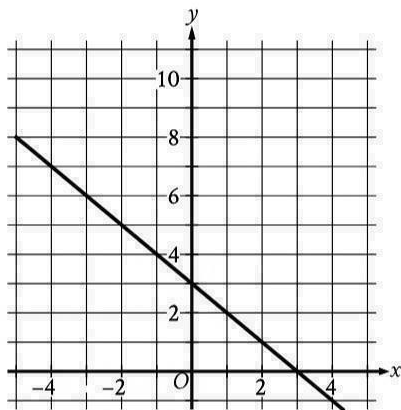
powered by
desmos

16

Mark for Review

The function q is defined by $q(x) = a|x - 9|^2 - 87|x - 9| + b$, where a and b are constants, and $a > b > 87$. If $q(639) = h$ and $q(-621) = k$, where h and k are constants, what is the value of $9(-639)^{h-k} + 621(9)^{k-h}$?

17.



The graph of the linear function $y = f(x) + 13$ is shown. If c and d are positive constants, which equation could define f ?

A. $f(x) = -d - cx$

B. $f(x) = d - cx$

C. $f(x) = d + cx$

D. $f(x) = -d + cx$



18



Mark for Review



The functions f and g are defined by the given equations below, where $x \geq 0$. Which of the following equations displays, as a constant or coefficient, the minimum value of the function it defines, where $x \geq 0$?

I. $f(x) = 16(1.3)^{x+5}$

II. $g(x) = 16(1.09)(1.3)^{x+3}$

- A. I only
- B. II only
- C. I and II
- D. Neither I nor II



19



Mark for Review



A set designer is hammering nails along the top edge of a wall with a length of $3x$ feet, where x is an integer, that will be used to hang garland. The designer hammers the first nail at the left edge of the wall. The designer then hammers an additional nail every $\frac{3}{8}$ feet along the entire length of the wall, starting from the first nail and with the last nail being hammered at the right edge of the wall. Which equation best represents this situation, where y is the number of nails the designer hammers along the entire length of the wall?

- A. $y = \frac{3}{8}x$
- B. $y = \frac{3}{8}x + 1$
- C. $y = 8x$
- D. $y = 8x + 1$



20



Mark for Review



The perimeter of an isosceles right triangle is $52 + 52\sqrt{2}$ inches. What is the length, in inches, of the hypotenuse of this triangle?

- A. 26
- B. $26\sqrt{2}$
- C. 52
- D. $52\sqrt{2}$



21



Mark for Review

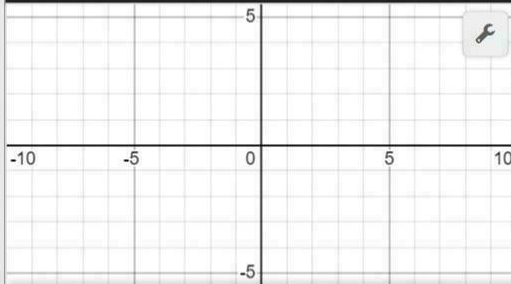


x	$g(x)$
-15	3
-6	0
9	5

The table shows three values of x and their corresponding values of $g(x)$, where $g(x) = \frac{f(x)}{x+3}$ and f is a linear function. What is the y -intercept of the graph of $y = f(x)$ in the xy -plane?

- A. (0,-6)
- B. (0,4)
- C. (0,8)
- D. (0,24)

Calculator



provided space, enter the decimal

provided space, enter it by

enter it as an improper fraction

a, or dollar sign.

OT receive credit

31/2

 $3 \frac{1}{2}$

powered by
desmos

2/3

.6666

.6667

0.666

0.66

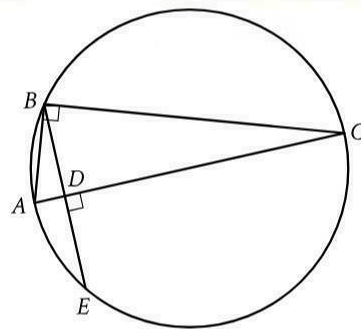
. 66

0.67

$$\frac{2}{3}$$

22

Mark for Review



Note: Figure not drawn to scale.

In the figure shown, points A , B , C , and E lie on the circle, and $AB < BC$. Segment AC is perpendicular to segment BE at point D , and $BD = \sqrt{226}$. The diameter of the circle is 115. If $\frac{CD}{AD} = r$, what is the value of r ?